Financial News Sentiment & Stock Price Analysis

A Production-Grade Data Science Portfolio Project

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Project: Week 1 Stock Challenge (Enhanced Week 12)

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Repository: https://github.com/Danielmituku/stock-challenge-week1

Executive Summary

This project analyzes the relationship between financial news sentiment and stock price movements for six major tech stocks. Originally developed in Week 1, it has been transformed in Week 12 into a production-grade portfolio piece demonstrating software engineering best practices valued by finance sector employers.

Key Achievements

Metric	Week 1	Week 12	Improvement
Code Structure	1 file (398 lines)	8 modules (1,200+ lines)	3x more organized
Test Coverage	0 tests	35 tests	inf improvement
Type Safety	0%	100% functions	Full type hints
Documentation	Basic README	Professional docs	Complete overhaul
Visualization	Static notebooks	Interactive dashboard	User-accessible

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1. Business Problem

The Challenge

Can we predict stock market movements by analyzing the sentiment of financial news?

Financial institutions spend billions on market research. Understanding the relationship between news sentiment and price movements could provide early warning signals for price changes, quantified sentiment metrics for trading decisions, and risk assessment tools for portfolio management.

Stocks Analyzed

Ticker	Company	Sector
AAPL	Apple Inc.	Technology
AMZN	Amazon.com	E-commerce/Cloud
GOOG	Alphabet Inc.	Technology
META	Meta Platforms	Social Media

MSFT	Microsoft Corp.	Technology
NVDA	NVIDIA Corp.	Semiconductors

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2. Solution Architecture

System Design

```
+--------------------------------------------------------------------------+
|                          DATA LAYER                                        |
+--------------------------------------------------------------------------+
|  News Data (1.4M headlines)          Stock Data (6 tickers)              |
|  +-- raw_analyst_ratings.csv         +-- AAPL, AMZN, GOOG, META,         |
|      - headline, date, publisher         MSFT, NVDA historical data      |
|      - stock ticker, URL                 - OHLCV daily prices            |
+--------------------------------------------------------------------------+
                                    |
                                    v
+--------------------------------------------------------------------------+
|                      PROCESSING LAYER (src/)                             |
+--------------------------------------------------------------------------+
|  src/data/                    src/analysis/          src/visualization/|
|  +-- loader.py                +-- sentiment.py       +-- plotting.py   |
|  |   - load_news_data()       |   - TextBlob                           |
|  |   - load_stock_data()      |   - VADER                              |
|  +-- preprocessor.py          +-- technical.py                         |
|      - clean_data()           |   - RSI, MACD, SMA                     |
|      - check_quality()        +-- statistics.py                        |
|                                   - correlation                        |
+--------------------------------------------------------------------------+
                                    |
                                    v
+--------------------------------------------------------------------------+
|                      PRESENTATION LAYER                                    |
+--------------------------------------------------------------------------+
|  Streamlit Dashboard (app/dashboard.py)                                  |
|  +-- Price Chart Tab      - Candlestick + indicators                     |
|  +-- Returns Tab          - Distribution + cumulative                    |
|  +-- Sentiment Tab        - Time series + pie chart                      |
|  +-- Data Table Tab       - Raw data exploration                         |
+--------------------------------------------------------------------------+
```

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3. Technical Implementation (Week 12 Improvements)

3.1 Code Refactoring

Before (Week 1): Single utility file with no type hints

```
# notebooks/utils.py (OLD - no type hints, mixed concerns)
def load_data(file_path='../data/raw.csv', sample_size=None):
    print("Loading dataset...")
    if sample_size:
        df = pd.read_csv(file_path, nrows=sample_size)
    else:
        df = pd.read_csv(file_path)
    return df
```

After (Week 12): Modular package with full type hints and dataclasses

```
# src/data/loader.py (NEW - typed, documented, validated)
from typing import Optional, Union
from pathlib import Path
import pandas as pd
from ..config import DataConfig

def load_news_data(
    file_path: Optional[Union[str, Path]] = None,
    sample_size: Optional[int] = None,
    config: Optional[DataConfig] = None,
) -> pd.DataFrame:
    """
    Load the raw analyst ratings (news) dataset.

    Parameters
    ----------
    file_path : str or Path, optional
        Path to the CSV file.
    sample_size : int, optional
        Number of rows to sample.
    config : DataConfig, optional
        Configuration object.

    Returns
    -------
    pd.DataFrame
        News data with columns: headline, date, publisher, stock

    Raises
    ------
    FileNotFoundError
        If the data file does not exist.
    """
    if config is not None:
        file_path = config.file_path
        sample_size = config.sample_size

    file_path = Path(file_path)
```

```
    if not file_path.exists():
        raise FileNotFoundError(f"Data file not found: {file_path}")

    # ... implementation
```

Configuration Dataclass:

```
# src/config.py
from dataclasses import dataclass, field
from typing import List, Optional

@dataclass
class TechnicalAnalysisConfig:
    """Configuration for technical analysis calculations."""

    rsi_period: int = 14
    rsi_overbought: float = 70.0
    rsi_oversold: float = 30.0
    macd_fast: int = 12
    macd_slow: int = 26
    macd_signal: int = 9
    sma_periods: List[int] = field(default_factory=lambda: [20, 50, 200])
    ema_periods: List[int] = field(default_factory=lambda: [12, 26])
    bollinger_period: int = 20
    bollinger_std: float = 2.0
```

3.2 Unit Testing Results

Test Coverage Summary:

Module	Tests	Pass Rate	Coverage
`src/config.py`	12	100%	95%
`src/analysis/sentiment.p	10	100%	88%
`src/analysis/technical.p	5	100%	92%
`src/analysis/statistics.	8	100%	90%
Total	**35**	**100%**	**91%**

Sample Test Output:

```
============================= test session starts ==============================
platform darwin -- Python 3.9.6, pytest-8.4.2
collected 35 items

tests/test_analysis.py::TestSentimentAnalysis::test_textblob_positive PASSED
tests/test_analysis.py::TestSentimentAnalysis::test_vader_negative PASSED
tests/test_analysis.py::TestTechnicalAnalysis::test_calculate_rsi PASSED
tests/test_analysis.py::TestTechnicalAnalysis::test_calculate_macd PASSED
tests/test_analysis.py::TestStatistics::test_gini_coefficient PASSED
tests/test_analysis.py::TestStatistics::test_correlation PASSED
tests/test_config.py::TestDataConfig::test_default_values PASSED
```

```
tests/test_config.py::TestDataConfig::test_invalid_config PASSED
... (27 more tests)

============================= 35 passed in 13.91s ==============================
```

3.3 Interactive Dashboard

Dashboard Screenshot Descriptions:

Tab 1: Price Chart with Technical Indicators

```
+------------------------------------------------------------------+
|  AAPL Stock Price with Technical Indicators                      |
+------------------------------------------------------------------+
|  [Candlestick Chart]                                             |
|  - Green/Red candles showing OHLC prices                         |
|  - Orange line: SMA 20-day                                       |
|  - Blue line: SMA 50-day                                         |
|  - Red line: SMA 200-day                                         |
|  - Gray band: Bollinger Bands (+/-2sigma)                              |
+------------------------------------------------------------------+
|  [RSI Chart - 0 to 100 scale]                                    |
|  - Purple line: RSI value                                        |
|  - Red dashed: Overbought (70)                                   |
|  - Green dashed: Oversold (30)                                   |
+------------------------------------------------------------------+
|  [MACD Chart]                                                    |
|  - Blue line: MACD                                               |
|  - Orange line: Signal                                           |
|  - Green/Red bars: Histogram                                     |
+------------------------------------------------------------------+
```

Tab 2: Returns Distribution

```
+---------------------------------+--------------------------------+
|  Daily Returns Distribution     |  Cumulative Returns            |
+---------------------------------+--------------------------------+
|  [Histogram]                    |  [Line Chart]                  |
|  - Bell curve shape             |  - Starting at 1.0             |
|  - Mean: 0.13% daily            |  - Ending at ~3.5x (250%)      |
|  - Std: 1.80%                   |  - Shows growth over time      |
|  - Range: -10% to +12%          |                                |
+---------------------------------+--------------------------------+
```

Tab 3: Sentiment Analysis

```
+---------------------------------+--------------------------------+
|  Sentiment Over Time            |  Category Distribution         |
+---------------------------------+--------------------------------+
|  [Scatter Plot]                 |  [Pie Chart]                   |
|  - X: Date                      |  - Green: Positive (29%)       |
|  - Y: Sentiment (-1 to 1)       |  - Red: Negative (19%)         |
```

```
|  - Color: Red/Yellow/Green     |  - Gray: Neutral (52%)         |
|  - Horizontal line at 0       |                                |
+------------------------------+------------------------------+
```

Key Metrics Panel:

```
+-------------+-------------+-------------+-------------+-------------+
| Latest Price | Daily Change | Volatility   | RSI          | Period Return|
|    $185.92   |    +1.23%    |    1.80%     | 58.4 Neutral |    +156.2%   |
+-------------+-------------+-------------+-------------+-------------+
```

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4. Key Results and Findings

4.1 Week 1 Analysis Results

Dataset Statistics:

Metric	Value
Total Headlines	1,407,328
Date Range	2011-2020
Unique Publishers	3,000+
Average Headline Length	73 characters
Valid Date Records	3.98%

Sentiment Distribution (55,230 analyzed headlines):

```
Sentiment Category Breakdown:
+-- Neutral:  52.13% ####################################################
+-- Positive: 28.66% ############################
+-- Negative: 19.20% ###################
```

Sentiment by Stock:

Stock	Mean Sentiment	Category
AMZN	+0.190	Most Positive
NVDA	+0.181	Very Positive
AAPL	+0.177	Very Positive
MSFT	+0.125	Positive
META	+0.098	Slightly Positive
GOOG	+0.017	Nearly Neutral

Technical Indicator Results:

Stock	Mean Daily Return	Volatility	Sharpe Ratio	Max Drawdown
NVDA	0.19%	2.89%	1.82	-18.76%
AAPL	0.13%	1.80%	1.45	-12.86%
AMZN	0.13%	2.18%	1.21	-14.05%
META	0.11%	2.53%	0.89	-26.39%

MSFT	0.10%	1.69%	1.18	-14.74%
GOOG	0.09%	1.73%	1.04	-11.10%

Correlation Analysis Limitation:

Issue	Impact
0.05-0.13% data overlap	Insufficient sample sizes
Missing timestamps	Cannot align news to pric
Lesson learned	Data alignment is critica

4.2 Week 12 Improvement Metrics

Code Quality Improvements:

Metric	Before	After	Improvement
Number of modules	1	8	8x more modular
Lines of code	398	1,247	Better organization
Functions with type hints	0	45	100% coverage
Documented functions	12	45	275% increase
Cyclomatic complexity (av	N/A	3.2	Low complexity

Testing Improvements:

Metric	Before	After
Unit tests	0	35
Test pass rate	N/A	100%
Code coverage	0%	91%
Test execution time	N/A	13.9s

Dashboard Capabilities Added:

Feature	Description	Business Value
Stock Selection	6 tickers available	Flexible analysis
Date Range Filter	Custom time windows	Historical comparison
Technical Indicators	RSI, MACD, SMA, EMA, BB	Trading signals
Sentiment View	Time series + distributio	News impact analysis
Data Export	Raw data table	Further analysis

5. Business Impact Assessment

Value for Finance Sector

Value Area	Benefit	Evidence
Risk Reduction	Calculation accuracy	35 unit tests, verified R
Maintainability	Easy updates	Modular code, type hints

Accessibility	Non-technical use	Interactive dashboard

Quantified Portfolio Value

Skill Demonstrated	Evidence	Finance Relevance
Software Engineering	8 modular packages	Production-ready code
Quality Assurance	35 tests, 91% coverage	Reliable calculations
Data Visualization	Interactive dashboard	Stakeholder communication
Documentation	Professional README	Team collaboration
Python Proficiency	Type hints, dataclasses	Modern best practices

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6. Limitations and Honest Assessment

Data Alignment Challenge

The correlation analysis faced a critical limitation:

```
News Data Coverage:  |####----------------| 3.98% valid dates
Stock Data Coverage: |####################| 100% trading days

Overlap for Analysis: |--------------------| 0.05-0.13%
```

Impact: Insufficient sample sizes (2-5 observations per stock) prevented statistically significant correlation conclusions.

Lesson Learned: Data preprocessing and alignment must be verified BEFORE conducting correlation analysis. This is a common real-world data science challenge.

What This Project Does NOT Prove

Statement	Explanation
Does NOT prove causation	News sentiment may correl
Does NOT provide trading	This is educational analy
Does NOT account for all	Market-wide events are no

What This Project DOES Demonstrate

Capability	Evidence
End-to-end workflow	Data loading to visualiza
Software engineering	Modular code, type hints,
Honest reporting	Limitations clearly docum
Production-ready code	CI/CD pipeline, 91% cover

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7. Future Improvements

If Given More Time

Priority	Improvement	Expected Impact
1	Obtain news data with bet	Enable proper correlation
2	Add SHAP explainability f	Increase transparency
3	Deploy dashboard to cloud	Public accessibility
4	Add real-time data feeds	Live analysis capability
5	Extend to more stocks/ind	Broader market coverage

Recommended Next Steps

```
Phase 1: Data Improvement
+-- Obtain news data with timestamps matching trading days

Phase 2: Model Development
+-- Train ML models using sentiment as features
+-- Add SHAP visualizations for model explainability

Phase 3: Production Deployment
+-- Deploy to cloud platform
+-- Add authentication and user management
+-- Implement real-time data streaming
```

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8. How to Run This Project

Quick Start

```
# Clone repository
git clone https://github.com/Danielmituku/stock-challenge-week1.git
cd stock-challenge-week1

# Create virtual environment
python -m venv .venv
source .venv/bin/activate  # Windows: .venv\Scripts\activate

# Install dependencies
pip install -r requirements.txt

# Run tests
pytest tests/ -v
```

```
# Launch dashboard
streamlit run app/dashboard.py
```

Project Structure

```
stock-challenge-week1/
+-- app/
|   +-- dashboard.py          # Streamlit dashboard
+-- src/
|   +-- config.py             # Configuration dataclasses
|   +-- data/                 # Data loading/preprocessing
|   +-- analysis/             # Sentiment, technical, statistics
|   +-- visualization/        # Plotting utilities
+-- tests/
|   +-- test_config.py        # 12 tests
|   +-- test_analysis.py      # 23 tests
+-- notebooks/                # Analysis notebooks
+-- INTERIM_REPORT.md         # Progress report
+-- FINAL_REPORT.md           # This document
+-- README.md                 # Project documentation
```

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9. Conclusion

This project demonstrates the journey from initial data analysis to a production-ready portfolio piece.

Category	Accomplishments
Technical Excellence	8 modular packages, 35 te
Professional Standards	Clean code, comprehensive
Finance Readiness	Verified calculations, ac
Honest Assessment	Data limitations identifi

Repository: https://github.com/Danielmituku/stock-challenge-week1

Dashboard: Run streamlit run app/dashboard.py

Tests: Run pytest tests/ -v (35 tests, all passing)

This report demonstrates production-grade data science practices including modular code design, comprehensive testing, interactive visualization, and honest communication of limitations-key competencies valued by finance sector employers.